YOUSSEF AMARIR

COMPUTER ENGINEERING | EMBEDDED SYSTEMS • ELECTRONICS • HARDWARE & FIRMWARE

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Portfolio: https://youssefamarir.github.io/

PROFESSIONAL SUMMARY

Computer Engineering candidate (UCF, December 2026) with 6+ years of enterprise IT experience and hands-on embedded firmware, PCB development, and electronics prototyping. Builds microcontroller systems in C/C++ with communication protocols, hardware testing, and strong troubleshooting.

EDUCATION

University of Central Florida | B.S. Computer Engineering, GPA 3.75 Expected Dec 2026

Coursework: Embedded Systems, FPGA/VHDL, Computer Architecture, Signals & Systems, Computer Networks.

TECHNICAL SKILLS

Programming / Firmware Embedded C, C++, Python, VHDL; bare-metal, FreeRTOS, interrupts, timers, ADC, PWM, state machines

Hardware / Validation Schematic capture, PCB layout, power regulation, board bring-up, soldering, oscilloscope, logic analyzer, multimeter, test plans

Microcontrollers / Interfaces STM32, MSP430, ESP32/ESP32-S3, Raspberry Pi; I2C, SPI, UART, CAN bus, GPIO, MQTT, Ethernet/TCP-IP

Development Tools STM32CubeIDE, PlatformIO, Code Composer Studio, ESP-IDF, Vivado, Git, Linux, SolidWorks

ENGINEERING PROJECTS

STM32 Real-Time CAN Bus Motor Control System | Independent Project | Embedded 2026

- Developed a two-node STM32G474 motor control system in embedded C, exchanging speed and direction commands over CAN bus.
- Integrated ADC input, interrupt-driven CAN reception, PWM actuation, and FreeRTOS task/queue communication; validated timeout and safe motor shutdown with hardware fault tests.

SENTRY - Environmental & Network Threat Monitoring | Independent & UCF Senior Design | Embedded Jan 2026 - Present

- Developed the ESP32-S3 sensor-node firmware and LVGL touchscreen interface, and built the Raspberry Pi hub (Python, Mosquitto, SQLite, PyQt5, Flask) for this UCF Senior Design Group 2 monitoring system. Source: github.com/youssefamarir/sentry-prototype
- Implemented BME688 I2C sensing, dual W5500 Ethernet and Wi-Fi, MQTT telemetry, and LLDP/CDP/FDP switch discovery; the final build adds an on-device camera with face unlock.
- Authored the software design, system testing, and executive-summary sections of the 127-page design report; PCB and hardware shared with the team.

Net Ghost - Handheld Network Diagnostics & Security Tester | Independent Project | Embedded 2026

- Built a handheld network tester on a Waveshare ESP32-S3-ETH with a round touchscreen (ESP-IDF, LVGL): on-device GUI plus a web interface served from the board's own Wi-Fi hotspot and a Bluetooth LE link.
- Implemented link/DHCP/gateway/DNS auto-testing, LLDP/CDP/FDP switch and VLAN discovery, live port statistics, and a packet analyzer with .pcap export over W5500 Ethernet and Wi-Fi.
- Added a read-only security audit suite (router exposure scan, Wi-Fi audit, rogue-AP and DNS-hijack checks, ARP-inspection test) for diagnosing vulnerabilities on an owned network. Source: github.com/youssefamarir/net-ghost

MSP430 Range Finder & Custom PCB | Individual Project | Embedded & PCB Design Aug - Dec 2025

- Designed and built a battery-powered ultrasonic range finder on an MSP430G2553: schematic and PCB layout in Fusion 360, plus custom 3.3 V and 5 V boost-regulator boards.
- Assembled the boards by stencil and reflow; verified the regulators under load with a multimeter, electronic load, oscilloscope, and thermal camera.
- Wrote the embedded C application for ADC-controlled PWM dimming, ultrasonic echo timing, and I2C LCD output. Source code: github.com/youssefamarir/msp430-range-finder

PROFESSIONAL EXPERIENCE

IT Field Support Specialist | Westgate Resorts, Orlando, FL Aug 2022 - Present

- Executed enterprise switch configuration and a full infrastructure replacement at the Ocoee building, supporting Cisco/Ruckus VLANs, uplinks, and port management.
- Built and diagnosed communications closets, tracing cabling, access-point, patch-panel, and network equipment faults across properties.
- Configured and tested endpoints using Active Directory, Entra ID, SCCM, Intune, BIOS tools, PowerShell, and Windows deployment workflows.
- Resolved production POS, serial-interface, payment-terminal, and print-server issues; received written recognition from three managers/directors.

IT Inventory Asset Management | Westgate Resorts, Orlando, FL Jul 2020 - Aug 2022

- Managed IT hardware inventory across properties; maintained asset records for procurement, refresh, and deployment planning.

Additional education: Valencia College, A.S. Computer Engineering, GPA 3.85 (2023). **Certification:** CompTIA A+.